

Calculating Social and Environmental Value

5

Overview

The previous chapters described the importance of balancing the various types of value; how that affects corporate governance; and how to discount future flows. With this chapter we take the necessary next step: how to calculate those types of value.

The core model in corporate finance is the discounted cash flow (DCF) model, used to determine the financial value (FV) of a project or a company. This chapter explains how social (S) and environmental (E) issues can be added to the standard DCF model. Recent advances in impact measurement enable companies to measure social and environmental quantities (such as life years saved by medical treatment or carbon emissions from using fossil fuels) and then to multiply these quantities by their respective shadow price, derived from welfare theory. The resulting value flows can be fed alongside the financial cash flows into the DCF model.

The practical challenge for calculating social value (SV) and environmental value (EV) is the availability of company information on S and E issues. Chapter 17 shows that companies are stepping up their sustainability reporting and that mandatory sustainability reporting standards are in the making. It is important to keep the big picture in mind by focusing on material S and E issues and not to get lost in unnecessary detail.

The reason for monetary valuation of (non-market priced) social and environmental impact is to make them visible, and part of the decision-making process, by integrating SV and EV in the accounting system and business language. A common unit (\$, € or any other currency) for financial, social, and environmental aspects of business impacts enables managers (and stakeholders) to compare different value components and to analyse the interactions between these value components. It facilitates better understanding of integrated value creation and incorporation of social and environmental impacts in strategy-setting and decision-making. Monetary valuation of impacts also allows managers to assess the relevance and materiality of sustainability topics. The mental challenge for many managers is to start thinking, analysing, and acting in this way, in spite of data gaps and other hurdles.

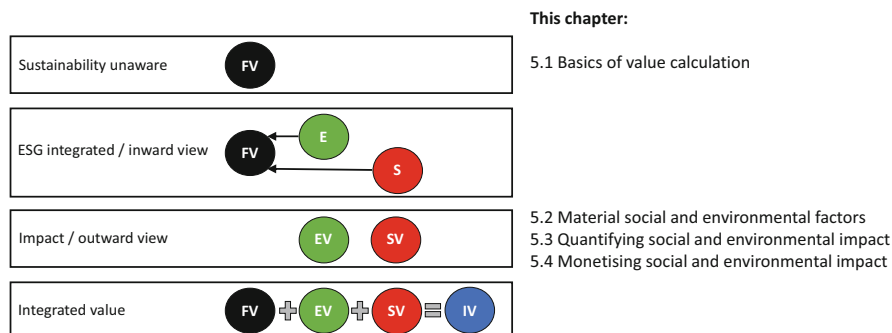


Fig. 5.1 Chapter overview

This chapter shows how to calculate FV (based on cash flows); SV (based on social value flows); and EV (based on environmental value flows) (Fig. 5.1). The next chapter analyses how the various value types can be used in investment decisions.

Learning Objectives

After you have studied this chapter, you should be able to:

- Use the discounted cash flow (DCF) model
- Calculate the social (S) and environmental (E) value of projects
- Assess the materiality of S and E factors
- Identify the advantages and shortcomings of the use of shadow prices

5.1 Basics of Value Calculation

The calculation of the value of a project or a company is at the heart of corporate finance. A commonly used model is the discounted cash flow (DCF) model, which derives the value V of a project or a company as follows:

$$V = \sum_{n=0}^N \frac{CF_n}{(1+r)^n} \quad (5.1)$$

whereby CF reflects the expected cash flows, r the discount rate (also called the cost of capital), and n the number of periods over which a cash flow is discounted.

The standard DCF model is used to calculate financial value FV . Chapter 6 shows how this is done for the value of projects, and Chap. 9 for the value of companies. This is the bedrock of corporate finance as found in current textbooks. This chapter shows how we can calculate social value SV and environmental value EV using the standard DCF model.

Social (S) and environmental (E) issues can be expressed in their own units Q (e.g. life years saved by medical treatment, or carbon emissions by using fossil fuels) and then multiplied by their respective shadow price SP derived from welfare

Table 5.1 Steel project

Year	2023	2024	2025	2026	2027	2028
Cash flows, in \$ millions	−100	40	40	40	40	40
Carbon emissions, in thousands tons	0	30	30	30	30	30

theory. The shadow price for one life year, for example, can be estimated at \$119,000 and the shadow price per 1 ton of CO₂ equivalent at \$224 (see Sect. A.1 in Appendix). The value flows VF are calculated as follows:

$$VF = Q \cdot SP \quad (5.2)$$

The social value flows SVF and environmental value flows EVF can be discounted with the DCF model to obtain SV and EV .

$$SV = \sum_{n=0}^N \frac{SVF_n}{(1+r)^n} \quad (5.3)$$

$$EV = \sum_{n=0}^N \frac{EVF_n}{(1+r)^n} \quad (5.4)$$

So, we obtain clear formulas to calculate SV and EV . Chapter 4 indicated that SV and EV should be discounted at the social discount rate, which is typically very low. The counterparty of companies' SV and EV is the wider society, representing current and future generations. Low social discount rates imply that current and future generations are treated as more or less equal. Here is a company example to show how the DCF model works for the various value types.

Company Example

Let's take a steel company that is considering a project to produce a new type of steel. This project employs a carbon-intensive technology and has a productive lifetime of 5 years. The financial discount rate is 6% per year and the social discount rate is 2% per year. Table 5.1 sets out the cash flow profile, and the carbon emissions, from the project. The project requires an initial investment of \$100 million and then yields \$40 million for the next 5 years. The carbon emissions of the production installation amount to 30,000 tons of CO₂ equivalent per year.

The first step is to calculate the environmental value flows. We use the earlier shadow carbon price of \$224 per 1 ton of CO₂ equivalent (that we assume to be constant over time) in Table 5.2. Applying Eq. (5.2), the resulting annual environmental value flow is −\$6.7 million (= 30,000 * \$224). Please note that this value is negative, because carbon emissions have a negative impact on the environment. Summing cash flows and environmental value flows provide us with the total value flows in the bottom line of Table 5.2.

The next step in Table 5.3 is to discount the cash flows and value flows at the discount factor $\frac{1}{(1+r)^n}$ (see Chap. 4). Let's start with the financial value. For year

Table 5.2 Value flows of steel project

Year	2023	2024	2025	2026	2027	2028
Cash flows, in \$ millions	–100	40	40	40	40	40
Carbon emissions, in thousands tons	0	30	30	30	30	30
Shadow carbon price, in \$	224	224	224	224	224	224
Environmental value flows, in \$ millions	0.0	–6.7	–6.7	–6.7	–6.7	–6.7
Total value flows, in \$ millions	–100.0	33.3	33.3	33.3	33.3	33.3

Table 5.3 Financial and environmental value components of steel project

Year	2023	2024	2025	2026	2027	2028
Cash flows, in \$ millions	–100	40	40	40	40	40
Discount factor, 6%	1.00	0.94	0.89	0.84	0.79	0.75
PV (cash flows), in \$ millions	–100.0	37.7	35.6	33.6	31.7	29.9
Financial value, in \$ millions	68.5					
Environmental value flows, in \$ millions	0.0	–6.7	–6.7	–6.7	–6.7	–6.7
Discount factor, 2%	1.00	0.98	0.96	0.94	0.92	0.91
PV (value flows), in \$ millions	0.0	–6.6	–6.5	–6.3	–6.2	–6.1
Environmental value, in \$ millions	–31.7					
Integrated value, in \$ millions	36.8					

1 (2024), the discount factor is $\frac{1}{(1.06)^1} = 0.94$, based on the financial discount rate of 6%. The present value PV of the value flows is obtained by multiplying the value flow with the discount factor. For 2024, the $PV = \$40 \text{ million} * 0.94 = \37.7 million . After adding up the present values PV for all years (2023–2028), you arrive at a financial value of \$68.5 million, as shown in Table 5.3.

Next, for the environmental value we use the social discount rate of 2%. For year 1 (2024), the discount factor is $\frac{1}{(1.02)^1} = 0.98$ and the $PV = -\$6.7 \text{ million} * 0.98 = -\6.6 million . Adding up for all years, the environmental value is $-\$31.7 \text{ million}$. The overall or integrated value is then $\$68.5 \text{ million} - \$31.7 \text{ million} = \$36.8 \text{ million}$ in Table 5.3.

What do our value calculations show? The steel project is worth doing, with a positive integrated value. The calculations also demonstrate that the carbon-intensive technology reduces the integrated value of the project by a significant amount. Management may want to consider an alternative technology which produces less carbon emissions (and thus a smaller negative environmental value).

The remainder of this chapter shows the detailed steps to calculating social value and environmental value. Chapter 6 develops investment decision rules for projects and the interaction between the various value types.

5.2 Material Social and Environmental Factors

The calculation of social value (SV) and environmental value (EV) is a recent phenomenon. This new field of impact measurement and valuation has developed methods to value social and environmental impact (Harvard Business School et al., 2022; Impact Economy Foundation, 2022; Impact Institute, 2019; Serafeim et al., 2019).

The value calculation for SV and EV can be done in three steps:

1. **Materiality assessment**—determine important SV and EV factors
2. **Quantification**—express these factors in their own units Q and
3. **Monetisation**—express these factors in money with shadow prices SP

This section discusses the materiality assessment to determine relevant social and environmental factors. Sections 5.3 and 5.4 explain the quantification and monetisation steps.

Materiality Assessment

Materiality assessments aim to determine which S and E factors are sufficiently important for consideration in SV and EV. Material social and environmental topics are those that reflect a company's most significant impacts (positive or negative) on people and environment. This is the outward impact of Fig. 2.5 in Chap. 2. Given the impact on a company's stakeholders and wider society, stakeholder engagement is crucial for companies to understand and determine materiality. Ultimately, companies decide which social and environmental topics should be included in their investment valuation calculations (Chaps. 6 and 7). However, this does not give them carte blanche, because they are likely to be held accountable for omissions. Moreover, new sustainability reporting rules prescribe certain sustainability topics on which it is mandatory to report (see Chap. 17). Box 5.1 discusses whether stakeholders have rights and could thus be seen as rightsholders.

Materiality depends on the specific situation and can differ per industry and country. For example, health and safety at work is a material topic for factories and mining operations. By contrast, attracting and training human talent is material for knowledge institutions, such as universities and management consultancies. In addition, materiality can differ within industries. Mining in the Democratic Republic of Congo has bigger human rights challenges than mining in Australia, for example.

Box 5.1: Stakeholders or Rightsholders?

The term stakeholder was popularised by Freeman (1984) and refers to a party that has an interest in a company and can either affect or be affected by the business. However, the sustainability platform R3.0 (2017, 2018) argues that 'rightsholder' is sometimes a better term to use. A rightsholder is a person or

(continued)

Box 5.1 (continued)

organisation that owns the legal rights to something. A right gives a much stronger claim than a stake. It recognises that people don't just have a desire for clean water (or other basics), but a right to it, giving them a stronger legal basis against company actions.

Such rights are laid down in international treaties and are increasingly referred to in legal proceedings, such as in the 2021 lawsuit against Royal Dutch Shell. In the ruling, the judge ordered the company to reduce its CO2 emissions by much more than the company intended to. Shell tried to hide behind its organisation in many local legal entities outside of the Netherlands, but the judge argued that headquarters effectively set the policy and strategy for those legal entities. This suggests that rightsholders might become more relevant than stakeholders.

In their 'Blueprint 1: Reporting', R3.0 (2017) put it as follows:

There is a need to strengthen and 'empower' rightsholders to remind organisations of their 'right to know' when it comes to duties and obligations, not allowing 'laissez-faire' as a widely used option of systemic malfunctioning. We argue for a pulling of the sustainability context and materiality principles together under the 'relevance' principle leading to the following steps, embedded in a plan-do-check-act approach for management.

Step 1: Identify impacts on capitals vital to rightsholder well-being: The first step in context-based materiality is to identify positive and negative company impacts on capitals (ecological, social, and economic resources) that are vital to rightsholder well-being. Companies have duties and obligations to uphold the well-being of their direct rightsholders, by managing their impacts on resources these rightsholders rely on.

Step 2: Determine if impacts compromise carrying capacities of capitals: The second step in context-based materiality is to determine if company impacts compromise the carrying capacity of capitals. If company impacts are far removed from this risk, then the impact can be deemed immaterial; if the impact is reasonably proximate to overshooting the carrying capacity of a capital, then it is by definition material.

Step 3: Ascertain strategic innovation opportunities to enhance capitals: The final step in context-based materiality is to ascertain if the impact lends itself to innovation opportunities with the potential to enhance or even regenerate capitals to achieve net positive impact.

The Impact Economy Foundation (2022) also takes a rights-based approach in the calculation of shadow prices (see Sect. 5.4 below).

Material Social and Environmental Factors

There is a core set of social and environmental factors, which one always needs to include in SV and EV calculations (Kuh et al., 2020):

- **Greenhouse gas emissions**, including carbon emissions
- **Labour practices**, including discrimination and inclusion
- **Business ethics**, including corruption and fraud

Table 5.4 provides an expanded set of social and environmental indicators that can be material for a company. Companies can use this larger set as a useful checklist in their materiality assessment. It should be noted that the concept of materiality is dynamic. Box 5.2 provides some examples of issues that have become more important over time.

Table 5.4 Material social and environmental factors

Factor	Example
Social factors	
<i>1. Labour practices</i>	
(a) Training	(a) New competences training
(b) Discrimination and inclusion	(b) Gender discrimination
(c) Health and safety (employees)	(c) Workplace health and safety
(d) Child labour and other human rights breaches	(d) Child labour in the value chain
(e) Employment well-being	(e) Additional benefits of employment
<i>2. Combatting poverty</i>	
(a) Underpayment in the value chain	(a) Paying below living wage
(b) Products/services that enable low-income people	(b) Microcredit
<i>3. Interaction with (local) communities</i>	
(a) Regional economic activity	(a) Local employment and suppliers
(b) Taxes	(b) Corporate tax
(c) Consumer well-being	(c) Consumer surplus of products sold
(d) Health and safety (local residents)	(d) Chemical plant safety
(e) Social cohesion	(e) Contributing to local sports club
(f) Business ethics	(f) Bribing (local) government officials
Environmental factors	
<i>1. Pollution</i>	
(a) Emissions of greenhouse gases	(a) Carbon emissions
(b) Toxic emissions to air	(b) Emission of particulate matter
(c) Toxic emissions to water	(c) Chemical spill
(d) Toxic disposition on land	(d) Nitrogen disposition
<i>2. Use of scarce resources</i>	
(a) Scarce materials	(a) Cobalt for batteries
(b) Land	(b) Deforestation for agriculture
(c) Water	(c) Water usage in production
<i>3. Restoration</i>	
(a) Air	(a) Removal of carbon emissions
(b) Land	(b) Land restoration
(c) Water	(c) Water purification

Source: Adapted and shortened from Impact Economy Foundation (2022)

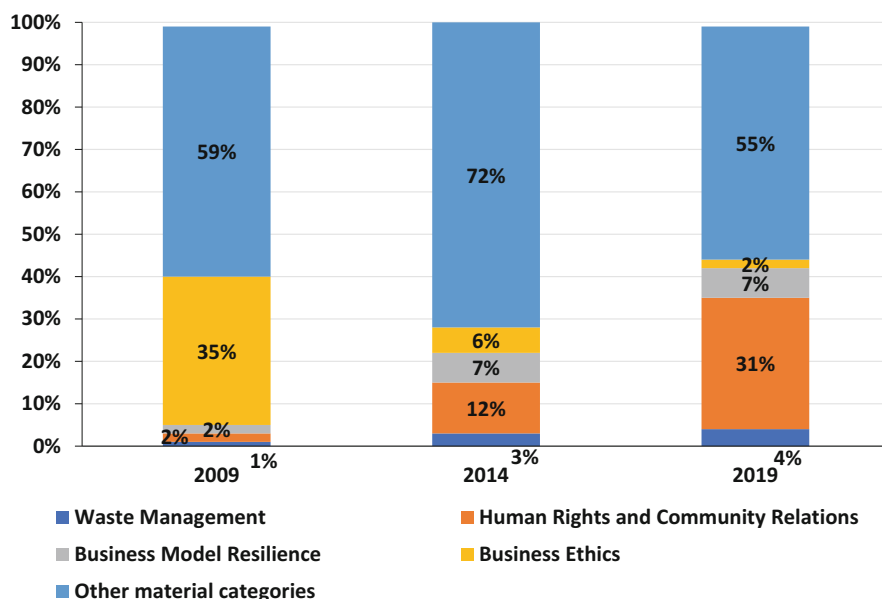


Fig. 5.2 Dynamic materiality in oil and gas industry. Source: Adapted from Kuh et al. (2020)

Box 5.2: Dynamic Materiality

Industry materiality is a dynamic concept. Certain topics that may not be considered material at one point may rise with respect to stakeholder focus over time. These shifts represent a change in the nature of what might be material to a company within a given industrial sector at a given time. An example is the topic of human rights and community relations in the oil and gas sector. That came to prominence in 2019, as witnessed by the legal case on Shell's treatment of indigenous people in Nigeria.¹ By contrast, business ethics has become less important over the same period, as illustrated in Fig. 5.2. The percentages in Fig. 5.2 represent the focus of stakeholder discussions in the oil and gas industry.

Examples 5.1 and 5.2 illustrate how the stakeholder impact map of Chap. 2 can be used to assess materiality in different industries. These examples confirm that material social and environmental factors are industry (and country) specific.

¹See <https://www.rechtspraak.nl/Organisatie-en-contact/Organisatie/Gerechtshoven/Gerechtshof-Den-Haag/Nieuws/Paginas/Shell-Nigeria-liable-for-oil-spills-in-Nigeria.aspx>.

Example 5.1: Determining Materiality in the Garment Industry

Problem

What are the material social and environmental factors in the garment (clothing) industry? In the garment industry, clothes are typically sold in high-income countries and produced in low-income countries. In the process, people in high-income countries have the benefits of cheap clothing, while the negative externalities tend to be imposed on people in low-income countries

Solution

A way to obtain clarity on the matter is to map the interests of stakeholders in a stakeholder impact map (see Chap. 2). For a company like Inditex or H&M, this could look as follows:

	Employees	Workers in the chain	Suppliers	Customers	Society at large
Stakeholder’s goals	Decent pay and working conditions	Decent pay and working conditions	Steady business	Cheap and fast fashion	Good S and E outcomes
Does the company help or hurt those goals?	Helps as it does provide the above	Hurts: poor conditions as costs are squeezed at suppliers	Hurts: unreliable partner that cancels orders after finishing	Helps: delivers that	Hurts: suffering in the chain (S) and large environmental damage (E)

Hence, there are serious frictions on both S and E between what customers want (cheap and fast fashion) and what matters to suppliers, workers in the chain and society at large. On S, material issues include labour conditions and supplier relations. After all, to keep prices low, fashion companies squeeze suppliers, who in turn squeeze their employees. On E, material issues include waste and GHG emissions. The high frequency of product replacement means that massive amounts of waste and GHG emissions are produced. Both S and E issues are at the heart of the business model of fast-fashion companies. However, not all companies are working sufficiently hard to address these challenges. ◀

Example 5.2: Determining Materiality in the Airline Industry

Problem

What are the material social and environmental factors in the traditional airline industry?

Solution

For a traditional airline like British Airways, Lufthansa, or American Airlines, the stakeholder impact map can look like this:

	Employees	Customers	Airports	Society at large
Stakeholder's goals	Decent pay and working conditions	Cheap tickets, good services, reliable, and safe transport	Fees, connections, traffic for retail operations	Jobs, tax income, access, environmental protection
Does the company help or hurt those goals?	Helps for pilots, hurts for other personnel: job cuts, long hours	Helps: delivers cheap tickets and safety; less so on reliability and service	Helps by running many flights	Hurts the environment (E) and health of residents near airports (S); delivers on most other aspects

This suggests serious friction between customers’ desire for cheap tickets and employees’ working conditions. Moreover, the negative externalities in health (S) and especially emissions (E) are substantial. For most airlines, negative value of the externalities probably outweighs their profitability by a wide margin. ◀

5.3 Quantifying Social and Environmental Impact

Let’s recall the three steps to calculate SV and EV:

- 1. **Materiality assessment**—determine important SV and EV factors
- 2. **Quantification**—express these factors in their own units *Q* and
- 3. **Monetisation**—express these factors in money with shadow prices *SP*

In this section, we analyse Step 2, quantification. This second step towards calculating SV and EV involves expressing S and E in their own units, similar to the volume component in sales or costs. For example, GHG emissions can be expressed in tonnes of CO₂ or tonnes of CO₂ equivalent, which includes all greenhouse gases (GHG): carbon dioxide CO₂ (80% of GHG), methane CH₄ (10% of GHG), nitrous oxide N₂O (7% of GHG), and fluorinated gases (3% of GHG). Table 5.5 does that for an airline, whose aircraft emit various GHG (expressed in CO₂ eq) by burning jet fuel (kerosene).

It should be noted, however, that this is just one component of E. To fully cover E, all nine planetary boundaries (see Chap. 1) should be considered and quantified as far as they are material for the company at hand. Some, such as nitrogen or freshwater use, are as easily quantifiable as carbon, but others are not. For example, biodiversity is difficult to express in a single metric, although some very

Table 5.5 Airline (partly) expressing E in its own units

Year	2022	2023	2024	2025	2026	2027	2028	2029	2030
CO ₂ -eq. million tons	30	31	22	25	31	30	29	25	20

general metrics such as mean species abundance (MSA) or hectares of land affected are used. Box 1.4 in Chap. 1 gives the drivers of biodiversity loss. Land-use change and land pollution—measured in hectares of land affected—are major drivers. The lack of an easily quantifiable metric is problematic since biodiversity loss is a major threat, at similar scale as climate change.

To provide an overview of the basics of environmental metrics, Sect. A.2 in the Appendix contains a primer on natural capital accounting. Ideally, every company would be held accountable for its contribution to all nine planetary boundaries and would have a budget of maximum harm that it is allowed to cause. This is the idea of thresholds and allocations proposed by R3.0 (2017). Current reporting is far from that ideal, but initiatives such as the European Sustainability Reporting Standards and the IFRS Sustainability Standards are moving reporting in that direction (see Chap. 17). Bolton et al. (2021) make recommendations to accomplish this for carbon budgets (see Box 5.3 below).

Box 5.3: Carbon Budgets

To stay within the realm of manageable temperature rises (i.e. below 2 °C warming), our carbon use cannot exceed a specific amount until 2050, by which time we need to be and remain carbon neutral. This effectively sets a global carbon budget, which is to be allocated over countries, industries, and companies, all of which will all need to establish an individual timeline for going to zero emissions. So far, however, this is hardly happening. Therefore, Bolton et al. (2021) recommend making carbon disclosure mandatory in the following way:

- Publicly listed firms are to report their global greenhouse gas emissions for the past calendar year in their annual reports. Private firms beyond a certain minimum size are to report their global greenhouse gas emissions for the past calendar year to a national registry in the country in which the firm is headquartered
- Corporate GHG emissions are expressed in tonnes of CO₂ equivalent, where the aggregation weights for greenhouse gases other than CO₂ are determined according to current IPCC guidelines
- Corporate GHG emissions comprise direct (scope 1) emissions from all installations and operating assets that the company (or its subsidiaries) has a majority interest in
- In addition to the above measure of gross direct carbon emissions (GDE), Bolton et al. (2021) support the reporting of corporate net direct carbon emissions (NDE), provided that GDE and NDE are reported separately. The NDE metric should only allow the subtraction from GDE of those carbon offsets that the firm, or its subsidiaries, has removed and sequestered durably from the atmosphere in the past year. Durability requires a

(continued)

Box 5.3 (continued)
reasonably high degree of confidence that the captured CO₂ will not be released back into the atmosphere for at least 100 years. That means that most of the current offsetting schemes will not be allowed

It is also possible to express components of S in their own units, such as quality life years added by a medical technology company (see Table 5.6). The number of quality life years added is calculated as the change in utility value induced by the medical treatment, multiplied by the duration of the treatment effect. The aim of the quality life year concept is to combine the biological, individual, and societal perspectives of health in a coherent fashion (Prieto & Sacristán, 2003). Section 5.4 discusses how the social and environmental quantities and shadow prices are based on welfare theory.

As with E, many S components are quite challenging to compute, since they lack a clear unit. This applies, for example, to violations of human rights. However, organisations like Impact Institute have shown that such types of S can still be quantified, albeit less easily. This can, for example, be done by expressing the components in terms of health effects or underpayment.

Attribution of Impact

Another challenge is attributing (i.e. distributing) shares of the impact to each of the stakeholders (Impact Economy Foundation, 2022). For example, if a construction company builds a windmill park, it cannot claim all of the positive (nor the negative) impacts of that project, since a lot of the impact is generated by the machinery companies that deliver the windmills and their components. Another example are carbon emissions from the usage of combustion engine vehicles. The emissions can be attributed to the car manufacturer (e.g. based on annual depreciation) and to the oil company selling petrol (e.g. based on the annual costs of petrol). This allows for an attribution of the emissions in proportion to responsibility. In the garment industry, for example, fashion companies have a shared responsibility for GHG emissions in their supply chain. In calculations, one can use the assumption that half of the GHG emissions by suppliers in the garment manufacturing can be attributed to the fashion company, as primary company in the supply chain (see Chap. 11 for the Inditex case study).

CO₂ emissions are probably the most widely used metric on the environmental side (natural capital). Several companies nowadays report on their scope 1, 2, and 3 emissions. The Greenhouse Gas Protocol (WRI, 2015) distinguishes between direct emissions from sources that are owned or controlled by the reporting entity;

Table 5.6 Medtech company (partly) expressing S in its own units

Year	2022	2023	2024	2025	2026	2027	2028	2029	2030
Quality life years added, × 1000	62	64	67	70	73	75	78	82	99

and indirect emissions that are a consequence of the activities of the reporting entity, but occur at sources owned or controlled by another entity. The GHG Protocol further categorises these direct and indirect greenhouse gas (GHG) emissions into three scopes:

- **Scope 1:** All direct GHG emissions of an organisation
- **Scope 2:** Indirect GHG emissions from consumption of purchased electricity, heat, or steam
- **Scope 3:** Other indirect GHG emissions both upstream and downstream of the value chain of an organisation

Scope 3 emissions are indirect emissions in the upstream supply chain caused by input purchases of the company, and by the use of the products sold by the company downstream. It is a challenge to attribute these indirect emissions across the value chain, without double counting (overreporting) or omission (underreporting).

While the Greenhouse Gas Protocol has a very clear definition for scope 3 emissions, it does not contain rules for attributing these emissions across the value chain (i.e. the supply chain). The Impact Economy Foundation (2022) proposes, in the case of shared responsibility, to attribute 50% of scope 3 emissions to the company with the prime responsibility (e.g. the car or garment manufacturer) and to re-attribute the remaining 50% over the value chain, based on how influential they are.

Example 5.3 shows how the social and environmental impact of a paint manufacturer can be attributed. The example shows that this paint manufacturer causes not only negative social and environmental impact in its own production process, but also in its supply chain. Scope 3 emissions form a major part of GHG emissions. We find the same for the fast-fashion retailer Inditex in Chap. 11.

Example 5.3: Attributing Impact

Problem

Akzo, a paint and coating manufacturer, has a sizeable social and environmental impact. From Akzo’s sustainability performance report 2021 (see Table 17.4 in Chap. 17), we take the following information:

People	Unit	2021
<i>People, process, and product safety</i>		
Fatalities employees	Number	1
Injury rate employees	/200k hours	0.21
Fatalities contractors	Number	0
Injury rate contractors	/200k hours	0.12

Planet	Unit	2021
<i>Energy use and emissions</i>		
GHG emissions—Scope 1	Kilotons	64.5
GHG emissions—Scope 2	Kilotons	172.1
Scope 3 upstream	Million tons	6.8
Scope 3 downstream	Million tons	7.7
<i>Resource efficiency</i>		
Freshwater use	Million m ³	9.6
Freshwater consumption	Million m ³	1.3

Please attribute the relevant social and environmental impact to Akzo.

Solution

On the social side, the fatalities and injury rate of Akzo's employees can be fully attributed to Akzo. Those of its contractors are a shared responsibility and are attributed for 50% to Akzo.

On the environmental side, scope 1 GHG emissions arise in Akzo's production process and are fully attributed to Akzo. These emissions amount to 64.5 kilotons, which is 0.0645 million tons. Scope 2 emissions occur at the electricity utility and Scope 3 emissions occur upstream and downstream in the supply chain. Scope 2 and 3 are attributed for 50% to Akzo, as explained in Sect. 5.3. The attributed scope 2 and 3 emissions amount to 7.3 million tons ($= 50\% * [0.172 + 6.8 + 7.7 \text{ million tons}]$). Interestingly, scope 1 emissions only form a minor part of overall GHG emissions: 0.9% ($= 0.065 / [0.065 + 7.3]$).

The freshwater usage of 10.9 million m³ is fully attributable to Akzo. ◀

The question of comparability arises: how to compare, say, GHG emissions with quality life years added? This is where monetisation comes in: by putting a unit price on these issues, they become comparable in monetary terms.

5.4 Monetising Social and Environmental Impact

Again, the three steps to calculate SV and EV are:

1. **Materiality assessment**—determine important SV and EV factors
2. **Quantification**—express these factors in their own units Q and
3. **Monetisation**—express these factors in money with shadow prices SP

In this section, we analyse the monetisation step, which refers to expressing S and E in monetary terms. This involves putting a price on the units identified in the second step. For example, for the above-mentioned airline example, Table 5.7 puts a price on carbon to arrive at the carbon component of EV. What price to choose is subject of debate: while the price of carbon in a certain market may be \$80 per tonne, estimates by scientists suggest that prices should be in the hundreds or even thousands to stay within our carbon budget (Boussemart et al., 2017). And the

Table 5.7 Airline (partly) monetising E

Year	2022	2023	2024	2025	2026	2027	2028	2029	2030
CO ₂ , million tons	30	31	22	25	31	30	29	25	20
Shadow carbon price, \$	224	232	240	248	257	266	275	285	295
Environmental value flow, \$ billions	-6.7	-7.2	-5.3	-6.2	-8.0	-8.0	-8.0	-7.1	-5.9

longer we wait, the higher the carbon price needs to be to reduce carbon emissions sharply to stay within our carbon budget. We reflect that in Table 15.7 by starting with a shadow carbon price of \$224 per tonne in 2022 (see Sect. A.1 in Appendix), which then rises with 3.5% every year (CE Delft, 2018; IEF, 2022).

Table 5.7 shows that the airline's GHG emissions of 30 million tons in 2022 translate to a negative environmental value flow of -\$6.7 billion. This is very substantial and can easily outweigh profit (i.e. the financial value flow), which is typically smaller for airlines of that size.

The Impact Economy Foundation (2022) defines impact as a change in capital (human, social, or natural capital), a change in experienced well-being (e.g. health effects) or a breach of a right (see Box 5.2 on rightsholders). Carbon emissions can be seen as a breach of the Paris Climate Agreement, which aims to limit global warming. The shadow carbon price for a tonne of CO₂ is then the price to restore the original situation, in this case taking one tonne of CO₂ out of the air. We explain below how shadow prices can be derived from welfare theory.

Similarly, Table 5.8 puts a price on the quality life years added by the aforementioned medical technology company. From a well-being perspective, the shadow price is \$119,000 per quality life year in 2022 and is assumed to be constant over time (see Sect. A.1 in Appendix). Table 5.8 illustrates that the annual social value flows of the medtech company can be sizeable, moving from \$7.4 billion in 2022 to \$11.8 billion in 2030.

As stated, other components of E and S are harder to quantify in their own units, but can nevertheless be monetised. For example, human rights violations can be expressed in monetary damages (by assessing how they hurt people's ability to lead a decent life) without taking the intermediate step of expressing them in comparable units.

Welfare-Based Shadow Prices

The shadow prices (also called monetisation factors by the Impact Economy Foundation) should reflect the 'true scarcity' of resources to stay within planetary boundaries; or the 'true price' of human rights breaches to stay within social boundaries. Using shadow prices is thus a tool for companies to stay within social and planetary boundaries, as discussed in Chaps. 1 and 2. The term shadow prices illustrates that these prices don't reflect current market prices but 'shadow' true prices. The Impact Economy Foundation (2022) and True Price (2021) provide a regularly updated list of impacts and shadow prices for a whole range of social and environmental impacts. Section A.1 in the Appendix provides a shortened list of

Table 5.8 Medtech (partly) monetising S

Year	2022	2023	2024	2025	2026	2027	2028	2029	2030
Quality life years, × 1000	62	64	67	70	73	75	78	82	99
Shadow price, × \$1000	119	119	119	119	119	119	119	119	119
Social value flows, \$ billions	7.4	7.6	8.0	8.3	8.7	8.9	9.3	9.8	11.8

shadow prices for illustration purposes. Box 5.4 shows that there are limits to monetisation—not everything can be quantified and monetised.

Box 5.4: Limits to Monetisation

The Capitals Coalition stresses that there are limits to monetisation. It may, for example, be difficult to monetise certain S issues (Social and Human Capital Coalition, 2019). Stakeholders may find it difficult to accept the quantification of certain changes (e.g. in cultural identity or historical significance). An example of the latter is provided by Rio Tinto in March 2020. The mining giant Rio Tinto destroyed a 46,000-year-old Aboriginal site in the expansion of an iron ore mine. One year later, the Rio Tinto chairman quit over the Aboriginal site damage and an Australian Parliamentary Inquiry ordered Rio Tinto to rebuild the ancient Aboriginal caves.

A precautionary approach may be needed for certain E issues (Natural Capital Coalition, 2016), for example when a company is close to important ecological thresholds (planetary boundaries) or has the potential to cause irreversible changes (e.g. species extinction). In these cases, the company should (aim to) avoid the use of these natural resources.

Rights

You may wonder about the theoretical underpinning of shadow or true prices for social and environmental impact. They are based on welfare theory (e.g. Bosselmann, 2016), whereby welfare is defined as the current and future value enjoyed by a company's stakeholders. True prices are based on two welfare categories: respect of rights and well-being. The first category of rights include (Galgani et al., 2021):

- **Human rights:** these refer to the rights of any individual as stated in the International Bill of Human Rights of the United Nations, such as the rights to life, liberty, and personal security, to freedom from slavery or degrading treatment
- **Labour rights:** these are the rights in the Fundamental Conventions of the International Labour Organisation, such as the rights to freely chosen work, to

fair wages, to a safe and healthy workplace, to unionise, and to freedom of discrimination

- **Environmental rights:** these refer to the right to a healthy environment and to natural resources, as enshrined in international agreements of the United Nations, such as the Paris Climate Agreement

In the latter case, for example, air, land, and water pollution and depletion of natural resources can be seen as breaches of environmental rights. The shadow price reflects the cost to restore the original situation or the cost to compensate for the damage by the unsustainable impacts.

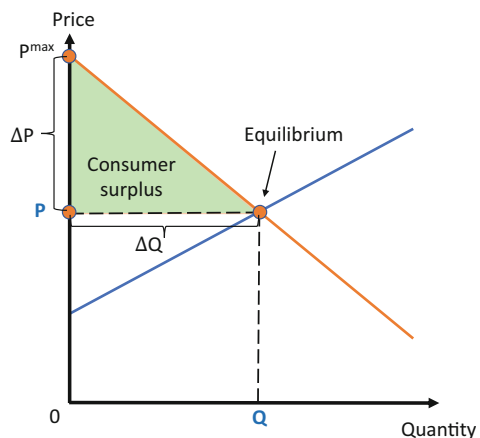
Well-Being

The second category is based on the well-being of stakeholders. Well-being, also known as quality of life, refers to what is intrinsically valuable for someone. This includes well-being of employees, customers, and communities (social cohesion). Employment well-being refers to additional well-being experienced by employees resulting from their employment and education at the company; this well-being is additional to the salary received. Employment well-being is measured by life satisfaction points on a scale of 0–100. The shadow price of one life satisfaction point is estimated at \$2647 (see Sect. A.1 in Appendix).

Consumer well-being is calculated as the consumer surplus, which is the difference between the price of a product and what consumers want to pay for it. Consumer surplus is a measure of consumer welfare. For completeness, we show how consumer surplus can be calculated. The shaded area below the downward sloping demand curve and above the equilibrium price in Fig. 5.3 is the consumer surplus:

$$\text{consumer surplus} = \Delta P \cdot Q \cdot \frac{1}{2} \quad (5.5)$$

Fig. 5.3 Consumer surplus



where the price differential ΔP is the maximum price P^{max} minus the price paid P ; and Q is the number of goods sold. Because ΔP cannot directly be observed, we use the price elasticity of demand. The price elasticity measures how demand $\Delta Q/Q$ reacts to a change in price $\Delta P/P$:

$$\text{price elasticity} = \frac{\Delta Q/Q}{\Delta P/P} \quad (5.6)$$

We can rewrite Eq. (5.6) as follows: $\Delta P = \frac{\Delta Q \cdot P}{\text{price elasticity} \cdot Q}$ and fill this expression into Eq. (5.5):

$$\text{consumer surplus} = \frac{\Delta Q \cdot P}{\text{price elasticity}} \cdot \frac{1}{2} = \frac{\text{sales}}{\text{price elasticity}} \cdot \frac{1}{2} \quad (5.7)$$

The numerator $\Delta Q \cdot P$ is equal to sales $Q \cdot P$, given that Fig. 5.3 shows that $\Delta Q = Q$. Equation (5.7) shows that a relatively high price elasticity yields a low consumer surplus (and vice versa).

We are now able to calculate the consumer surplus. The only input required is an estimate of the price elasticity. In the case of Inditex in Chap. 11, Khaled and Lattimore (2006) find an average price elasticity of men's and women's clothing of 3.45. Given Inditex sales of €20.4 billion, the estimate of the consumer surplus amounts to €3.0 billion (= €20.4 billion/3.45 * 0.5).

The purpose of these calculations is to show that employment and consumer well-being can be estimated. Standard microeconomic tools can be used to make the calculations. Of course, estimates of the degree of life satisfaction (measured in life satisfaction points) and the price elasticity of demand need to be made. In both cases, well-being is defined as the 'benefits' on top of the financial payments—salaries paid to employees and market prices paid by consumers for goods and services.

Calculating Social and Environmental Value

With the outputs from all three steps, we can calculate the value flows VF from Eq. (5.2):

$$VF = Q \cdot SP$$

The social value flows SVF and environmental value flows EVF in Tables 5.7 and 5.8 can subsequently be discounted with the standard DCF model to obtain the social value SV and environmental value EV of a project or a company. Equations (5.3) and (5.4) provide the DCF model for SV and EV :

$$SV = \sum_{n=0}^N \frac{SVF_n}{(1+r)^n}$$

$$EV = \sum_{n=0}^N \frac{EVF_n}{(1+r)^n}$$

Table 5.9 Environmental value (EV) of airline (in \$ billions)

Year	2022	2023	2024	2025	2026	2027	2028	2029	2030
Environmental value flows (EVF)	-6.7	-7.2	-5.3	-6.2	-8.0	-8.0	-8.0	-7.1	-5.9
Discount factor, 2%	1	0.98	0.96	0.94	0.92	0.91	0.89	0.87	0.85
PV (EVF)	-6.7	-7.0	-5.1	-5.9	-7.4	-7.2	-7.1	-6.2	-5.0
Environmental value (EV)	-57.6								

Table 5.10 Social value (SV) of medtech (in \$ billions)

Year	2022	2023	2024	2025	2026	2027	2028	2029	2030
Social value flows (SVF)	7.4	7.6	8.0	8.3	8.7	8.9	9.3	9.8	11.8
Discount factor, 2%	1	0.98	0.96	0.94	0.92	0.91	0.89	0.87	0.85
PV (SVF)	7.4	7.5	7.7	7.8	8.0	8.1	8.2	8.5	10.1
Social value (SV)	73.3								

We apply the social discount rate of 2% to discount social and environmental value flows, as discussed in Chap. 4. Table 5.9 shows the outcome. The environmental value flows *EVF* are multiplied with the discount factor to obtain the present value of *EVF*. When we sum the present values *PVs* of *EVF* we get the environmental value *EV*, which is -\$57.6 billion for the airline. So, our airline has a large negative *EV*.

Table 5.10 follows the same procedure to calculate the social value *SV* of the medtech company. Our medtech appears to achieve a large positive *SV* of \$73.3 billion.

Application in Case Studies

The best way to understand the working of shadow prices in the calculation of social and environmental value is to apply them in company case studies. Chapter 11 applies shadow prices to calculate the integrated value of Inditex and Chapter 18 uses shadow prices to assess the (dis)synergies from the aborted Kraft Heinz–Unilever takeover. These case studies illustrate the importance of valuing social and environmental impacts with shadow prices. The resulting social and environmental value can be larger than the company's financial value.

5.5 Conclusions

The core model in corporate finance is the discounted cash flow (DCF) model to determine the financial value (FV) of a project or a company. This chapter explained how social (S) and environmental (E) issues can be added to the standard DCF

model. Recent advances in impact measurement enable companies to measure social and environmental quantities, such as life years saved or carbon emissions, and then to multiply these quantities by their respective shadow price, derived from welfare theory. The resulting value flows can be put, alongside the financial cash flows, into the DCF model.

The challenge for calculating social value (SV) and environmental value (EV) is the availability of company information on S and E issues. Chapter 17 shows that companies are stepping up their sustainability reporting and that mandatory sustainability reporting standards are in the making. It is important to keep the big picture by focusing on material S and E issues, and not to get lost in unnecessary detail.

This chapter showed how to calculate FV (based on cash flows), SV (based on social value flows), and EV (based on environmental value flows). The next chapter analyses how the various value types can be used in investment decisions.

Key Concepts Used in This Chapter

Attribution of impact refers to attributing or distributing shares of the impact to each of a company's stakeholders

Impact is defined as a change in capital (human, social, or natural capital), a change in experienced well-being or a breach of a right

Integrated value is obtained by combining the financial, social, and environmental values in an integrated way (with regard for the interconnections)

Materiality indicates relevant and significant information

Materiality assessment aims to determine which S (social) and E (environmental) factors are sufficiently important for consideration in SV and EV

Monetisation of social and environmental factors means to express them in monetary terms with shadow prices

Quantification of social and environmental factors means to express them in their own units

Rights refer to human, labour, and environmental rights of individuals as laid down in international treaties

Rightholder is a person or organisation that owns the legal rights to something

Shadow prices or true prices reflect the 'true scarcity' of resources to stay within planetary boundaries or the 'true price' of human right breaches to stay within social boundaries; shadow prices are based on welfare theory

Stakeholder refers to a person or organisation that has an interest or 'stake' in the company: customers, employees, suppliers, shareholders, creditors, and the community

True prices, see shadow prices

Well-being or quality of life refers to what is intrinsically valuable for someone

Welfare is current and future value enjoyed by stakeholders

Appendix: Shadow Prices and Natural Capital Accounting

This Appendix provides a list of shadow prices and additional material on natural capital accounting, as discussed in Sect. 5.4.

A.1 List of Shadow Prices

The Impact Economy Foundation (2022) and True Price (2021) publish a regularly updated list of shadow prices (or monetisation factors) to monetise social and environmental impact. Section 5.4 discusses the theoretical underpinning of these shadow prices from welfare theory. In this Appendix, we show some commonly used shadow prices for the year 2022 for illustration purposes. In addition, there are, for example, several shadow prices for air, land, and water pollution (in the form of toxic emissions). See the guidance document for the full list:

<https://impacteconomyfoundation.org/impactweightedaccountsframework/> and <https://trueprice.org/monetisation-factors-for-true-pricing/>

Impact	Indicator	Shadow price	Explanation
<i>Environmental impacts</i>			
Climate change	GHG emissions	\$224/ton CO ₂ equivalent (eq)	A restoration cost that expresses the abatement cost for achieving the policy targets of reducing GHG emissions to meet the 2 °C target of the Paris Agreement
Air pollution	Toxic emissions to air	\$119,000/DALY (disability-adjusted life year) ^a	A compensation cost that expresses the Value of Statistical Life (VSL) based on a meta-analysis of willingness-to-pay studies
	Nitrogen deposition NH3 (animal husbandry)	\$18.10/kg NH3 eq	A marginal cost of the abatement measures needed to reach the regulatory target of nitrogen deposition in nature areas
	Nitrogen deposition NOx (use of machines and vehicles)	\$1.76/kg NOx eq	
	Particulate matter (PM) formation	\$75/kg PM2.5 eq	A compensation cost that expresses the social cost of pollution and indicates the occurring loss of economic welfare when pollutants are emitted into the environment, looking at human health damage and ecosystems damage
	Photochemical oxidant formation (POF)	\$1.18/kg NMVOC \$4.19/kg NOx eq	
	Ozone layer depleting emissions	\$65.40/kg CFC-11 eq	

(continued)

Impact	Indicator	Shadow price	Explanation
Water pollution	Toxic emissions to water	\$119,000/DALY (disability-adjusted life year)	A compensation cost that expresses the Value of Statistical Life (VSL) based on a meta-analysis of willingness-to-pay studies
	Freshwater eutrophication ^b	\$290/kg phosphorus eq to freshwater	A combination of restoration and compensation costs based on a literature review on the costs of eutrophication. Restoration costs express average abatement costs for bringing nutrient levels to a regulatory target, for the impacts that are reversible
Soil pollution	Toxic emissions to soil	\$119,000/DALY (disability-adjusted life year)	A compensation cost that expresses the Value of Statistical Life (VSL) based on a meta-analysis of willingness-to-pay studies
	Terrestrial ecotoxicity	\$0.4/ton 1,4 dichlorobenzene (DB) emitted to industrial soil eq	A compensation cost that expresses the social cost of pollution and indicates the occurring loss of economic welfare when pollutants are emitted into the environment, looking at ecosystems damage
	Freshwater ecotoxicity	\$57.90/ton 1,4-DB emitted to freshwater eq	
Soil degradation	Soil organic carbon (SOC) loss	\$43/ton SOC loss	A compensation cost that expresses the damage cost for the chemical, physical, biological, and ecologic decline of soil resulting from loss of soil organic carbon

(continued)

Impact	Indicator	Shadow price	Explanation
Land occupation	Tropical forest	\$3030/(MSA * ha * year)	A compensation cost that expresses the opportunity cost of land occupation based on the value of ecosystem services for main biomes
	Other forest	\$1450/(MSA * ha * year)	
Availability of non-renewable materials	Non-renewable material depletion	\$261/ton copper eq	A compensation cost that expresses the future loss of economic welfare resulting from increased extraction costs of non-renewable materials in the future
Availability of water	Scarce blue water use	\$1.49/m ³	A restoration cost that expresses the annualised cost of desalination, including the cost of operation and maintenance, electrical and thermal energy, as well as the cost of covering and repaying initial capital and operational costs of desalination
<i>Social impacts</i>			
Effects on human health	Effects on human health	\$119,000/DALY (disability-adjusted life year)	A compensation cost that expresses the Value of Statistical Life (VSL) based on a meta-analysis of willingness-to-pay studies
Consumer well-being	Consumer surplus	\$ based on price elasticity of demand	The value of well-being is based on the consumer surplus. See Sect. 5.4
Well-being of employment	Well-being effect per one additional point of life satisfaction	\$2647/life satisfaction point (scale 0–100)	The value of well-being is based on a reduction from unemployment and an increase of well-being resulting from education

(continued)

Impact	Indicator	Shadow price	Explanation
Occupational health and safety incidents	Non-fatal occupational incidents	\$4170/incident	A combination of compensation, prevention, and retribution costs. The compensation cost represents the average cost of medical expenses for occupational injuries not covered by the employer. The prevention cost expresses the cost of generic auditing set-up to prevent future instances. Finally, the retribution costs represent a penalty for the cases in which workers perform their duties in conditions that violate health and safety regulations
	Fatal occupational incidents	\$3,540,000/incident	
	Occupational injuries with breach of health and safety standards	\$3840/incident	
Underpayment in the value chain	Wage gap of workers earning below minimum wage	\$1.56/\$	A combination of compensation, prevention, and retribution costs. The compensation cost expresses the gap to a decent living wage, as well as the interest rate. The prevention cost expresses the cost of generic auditing set-up to prevent future instances. The retribution cost represents a penalty for the wage gap that is below the legal minimum wage
	Wage gap of workers earning above minimum wage but below decent living wage	\$1.06/\$	

(continued)

Impact	Indicator	Shadow price	Explanation
Child labour	Workers below minimum age for light work involved in non-hazardous economic work	\$21,600/child FTE	A combination of restoration, compensation, prevention, and retribution costs. The restoration cost expresses the costs of providing quality education for children not attending school and
	Underage workers above minimum age for light work and below minimum age involved in non-hazardous light economic work	\$7970/child FTE	the costs of implementing additional components of reintegration programmes for children involved in hazardous child labour. The compensation cost expresses the loss of future earnings when a child is prevented from attending school during youth. The prevention cost expresses the cost of generic auditing set-up to prevent future instances. Finally, the retribution cost represents a penalty for instances of child labour
	Underage workers who are not attending school	\$25,300/children	
Forced labour	Forced workers	\$17,200/FTE	A combination of restoration, compensation, prevention, and retribution costs. The restoration cost expresses the restitution of past economic losses of forced workers in debt bondage, as well as other costs for reintegration. The compensation cost expresses the cost of lost health valued using DALY for forced workers as victims of abuse. The prevention cost expresses the cost of generic auditing set-up to prevent future instances

(continued)

Impact	Indicator	Shadow price	Explanation
Discrimination	Female workers without provision for maternity leave	\$2450/FTE	A combination of restoration, prevention, and retribution costs. The restoration cost represents the restitution of wage lost due to denied maternity leave, gender discrimination, and unequal opportunities. The prevention cost expresses the cost of generic auditing set-up to prevent future instances of discrimination. The retribution cost represents a penalty for the violation of denied maternity leave and a penalty proportional to the size of the wage gap from discrimination
	Wage gap from gender discrimination	\$1.06/\$	
	Wage gap from unequal opportunities	\$1.06/\$	
Occurrence of harassment	Workers who experienced non-physical non-sexual harassment	\$33,000/worker	A combination of restoration, compensation, prevention, and retribution costs. The restoration cost represents average medical costs for injuries, anxiety, depression, and post-traumatic stress disorder resulting from workplace harassment. The compensation cost represents the cost of loss of future well-being resulting from long-term mental health impact of victims of harassment. The prevention cost expresses the cost of generic auditing set-up, to prevent future instances. Finally, the retribution cost represents a penalty for instances of physical non-sexual and sexual harassment based on the weighted average of penalties from various countries to express a global penalty
	Workers who experienced non-physical sexual harassment	\$35,700/worker	
	Workers who experienced physical non-sexual harassment	\$64,300/worker	
	Workers who experienced non-severe physical sexual harassment	\$74,500/worker	
	Workers who experienced severe physical sexual harassment	\$85,800/worker	

(continued)

Impact	Indicator	Shadow price	Explanation
Lack of freedom of association	Instances of denied freedom of association	\$527/violation	A combination of prevention and retribution costs. The prevention cost expresses the cost of generic auditing set-up to prevent future instances. The retribution cost expresses a penalty for denied freedom of association (e.g. to form a trade union)

Source: Adapted and shortened from Impact Economy Foundation (2022)

^aDisability-adjusted life year (DALY) combines (1) years of life lost due to premature mortality; and (2) years of life lost due to time lived in states of less than full health, or years of healthy life lost due to disability. One DALY represents the loss of the equivalent of one year of full health

^bEutrophication is the process by which an entire body of water, or parts of it, becomes progressively enriched with minerals and nutrients, particularly nitrogen and phosphorus

A.2 Natural Capital Accounting

Natural capital accounting provides an accounting framework to measure stocks and flows of natural capital (Hoekstra, 2022). The underlying premise: because the environment is important to society and the economy, it should be recognised as an asset. It must, therefore, be maintained and managed, with its contributions (services) better integrated into commonly used frameworks like the System of National Accounts, which defines important economic variables such as GDP. The System of Environmental-Economic Accounting (SEEA) is the accepted international standard for environmental-economic accounting, providing a framework for organising and presenting statistics on the environment and its relationship with the economy. It brings together economic and environmental information in an internationally agreed set of standard definitions, classifications, and accounting rules to produce internationally comparable statistics.

The SEEA is developed and released under the auspices of the United Nations, International Monetary Fund and World Bank. It consists of two parts. The **SEEA Central Framework** was adopted as the international standard for environmental-economic accounting in 2012. The Central Framework looks at ‘environmental assets’, such as water resources, energy resources, forests, and fisheries. It considers their use in the economy and returns to the environment in the form of waste, air, and water emissions. In addition, there are methodological documents that take a sectoral approach, such as SEEA-Energy; SEEA-Water and the SEEA Agriculture, Forests and Fisheries. The **SEEA Ecosystem Accounting** complements the Central

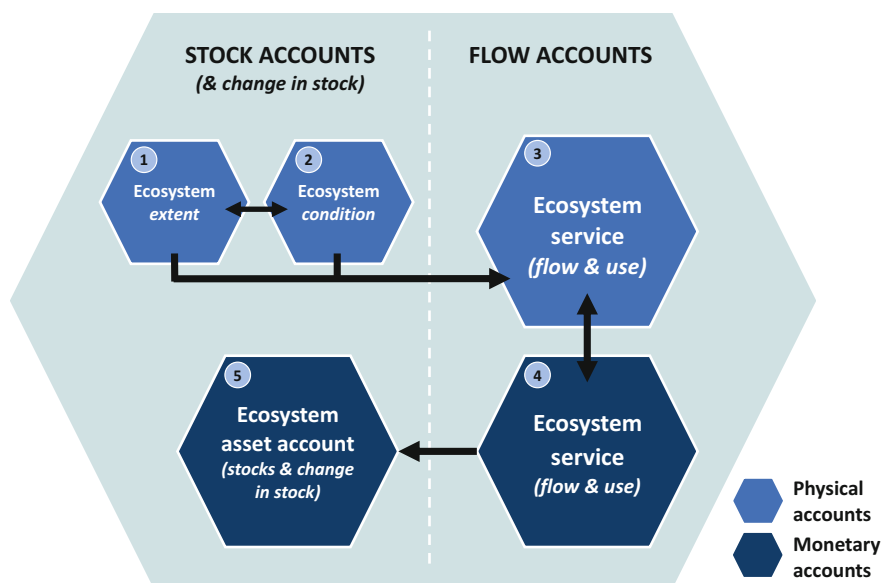


Fig. 5.4 SEEA—sequence of ecosystem accounts. Source: Adapted from SEEA, UN

Framework and was adopted in 2021. It takes the perspective of ecosystems and considers how individual environmental assets interact as part of natural processes within a given spatial area. Ecosystem accounts enable the presentation of indicators of the level and value of ‘ecosystem services’ in a given spatial area. The SEEA Ecosystem Accounts consist of five different types of accounts, depicted in Fig. 5.4:

1. **Ecosystem Extent** accounts record the total area of each ecosystem, classified by type within a specified area (ecosystem accounting area). Ecosystem extent accounts are measured over time in ecosystem accounting areas (e.g., nation, province, river basin, protected area, etc.) by ecosystem type, thus illustrating the changes in extent from one ecosystem type to another over the accounting period.
2. **Ecosystem Condition** accounts record the condition of ecosystem assets in terms of selected characteristics at specific points in time. Over time, they record the changes to their condition and provide valuable information on the health of ecosystems.
3. & 4. **Ecosystem Services** flow accounts (physical and monetary) record the supply of ecosystem services by ecosystem assets and the use of those services by economic units, including companies and households.
5. **Monetary Ecosystem Asset** accounts record information on stocks and changes in stocks (additions and reductions) of ecosystem assets. This includes accounting for ecosystem degradation and enhancement.

For more on natural capital accounting: <https://seea.un.org/content/seea-e-learning-resources>.

Suggested Reading

Harvard Business School, Impact Economy Foundation, Value Balancing Alliance, and Others. (2022). *The case of monetary valuation*.
 Impact Economy Foundation (IEF). (2022). *Impact-weighted accounts framework*.
 Serafeim, G., Zochowski, R., & Downing, J. (2019). *Impact-weighted financial accounts: The missing piece for an impact economy*. White Paper, Harvard Business School.

References

- Bolton, P., Reichelstein, S., Kacperczyk, M., Leuz, C., Ormazabal, G., & Schoenmaker D. (2021). Mandatory Corporate Carbon Disclosures and the Path to Net Zero. *Management and Business Review*, 1(3), 21–28.
- Bosselmann, K. (2016). *The principle of sustainability: transforming law and governance* (2nd ed.). Routledge.
- Boussemart, J., Leleu, H., & Shen, Z. (2017). Worldwide carbon shadow prices during 1990–2011. *Energy Policy*, 109, 288–296.
- CE Delft. (2018). Environmental prices handbook; EU28 version, Delft.
- Freeman, R. E. (1984). *Strategic management: A stakeholder approach*. Cambridge University Press.
- Galgani, P., Woltjer, G., de Adelhart Toorop, R., & de Groot Ruiz, A. (2021). Valuation framework for true price assessment of agri-food products, version 1. True Price and Wageningen University & Research.
- Harvard Business School, Impact Economy Foundation, Value Balancing Alliance, and Others. (2022). *The case of monetary valuation*.
- Hoekstra, R. (2022). *How natural capital accounting can help accelerate finance for nature*. United Nations.
- Impact Economy Foundation (IEF). (2022). *Impact-weighted accounts framework*.
- Impact Institute. (2019). *Framework for impact statements (FIS)*.
- Khaled, M., & Lattimore, R. (2006). The changing demand for apparel in New Zealand and import protection. *Journal of Asian Economics*, 17(3), 494–508.
- Kuh, T., Shepley, A., Bala, G., & Flowers, M. (2020). *Dynamic materiality: Measuring what matters*. Truvalue Labs.
- Natural Capital Coalition. (2016). Natural Capital Protocol.
- Prieto, L., & Sacristán, J. (2003). Problems and solutions in calculating quality-adjusted life years (QALYs). *Health and Quality of Life Outcomes*, 1(1), 1–8.
- R3.0. (2017). Blueprint 1: Reporting. A principles-based approach to reporting serving a green, inclusive & open economy, R3.0 Blueprint series.
- R3.0. (2018). Blueprint 5: The transformation journey—A step-by-step approach to organizational thriveability and system value creation, R3.0 Blueprint series.
- Serafeim, G., Zochowski, R., & Downing, J. (2019). *Impact-weighted financial accounts: The missing piece for an impact economy*. White Paper, Harvard Business School.
- Social and Human Capital Coalition. (2019). Social & human capital protocol.
- True Price. (2021). Monetisation factors for true pricing, version 2.0.3.
- World Resources Institute (WRI). (2015). Greenhouse gas protocol.

Open Access This chapter is licensed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>), which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license and indicate if changes were made.

The images or other third party material in this chapter are included in the chapter's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the chapter's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

